

4(a) Electric field strength at a point is electric force per unit positive charge at that point. [1]

(b)(i) $F = (2e)E$ [1]

$$= 2 \times (1.6 \times 10^{-19}) (7.5 \times 10^4)$$

$$= 2.4 \times 10^{-14} \text{ N} \quad [1]$$

(b)(ii) Time taken for alpha particles to travel 1 m in the horizontal direction,

$$t = \frac{s_x}{u_x} = \frac{1.0}{1.50 \times 10^7}$$

$$= 6.67 \times 10^{-8} \text{ s} \quad [1]$$

(b)(iii) Acceleration in the vertical direction,

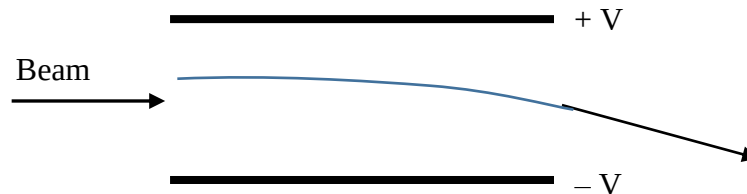
$$a = \frac{F}{m} = \frac{F}{4u} = \frac{2.4 \times 10^{-14}}{4 \times 1.66 \times 10^{-27}} = 0.3614 \times 10^{13} \text{ ms}^{-2} \quad [1]$$

Displacement in vertical-direction during time t,

$$s = \frac{1}{2}at^2 = \frac{1}{2}(0.3614 \times 10^{13})(6.67 \times 10^{-8})^2 = 0.008039 \approx 0.0080 \text{ m} \quad [1]$$

The particles will not hit any of the plates as the vertical displacement of the electron is less than 0.0125 m when it is travelling between the two parallel plates.

(b)(iv)



Parabolic path curves downward inside the plates

Straight path outside the plates